

E-Commerce Analytics Report

Olist Dataset · BigQuery SQL Analysis · 2016–2018

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100K+

Orders analyzed

9

Relational tables

6

Business questions

2016–2018

Data period

This report presents a comprehensive SQL analysis of the Olist Brazilian e-commerce dataset, answering 6 critical business questions across revenue, customer behavior, logistics, satisfaction, and seller quality. All queries were built in Google BigQuery using advanced SQL including window functions, CTEs, RFM segmentation, and conditional aggregations.

01 Monthly GMV & Month-over-Month Growth

SQL TECHNIQUES: `DATE_TRUNC` · `LAG()` · Window functions · % variance

Key Finding

The platform showed consistent GMV growth from late 2016 through mid-2018, with several months exceeding 40% month-over-month growth. A dip in April and June 2017 reveals seasonal patterns. December 2016 was excluded from growth calculations due to incomplete data (\$10.9 GMV), likely reflecting the platform's soft launch period.

02 Revenue Pareto by Product Category

SQL TECHNIQUES: `SUM() OVER()` cumulative · % of total · Pareto filter

Key Finding

Just 16 out of 70+ categories account for 80% of total revenue. The top 3 categories — `beleza_saude` (\$1.26M), `relorios_presentes` (\$1.21M), and `cama_mesa_banho` (\$1.04M) — represent over 26% of GMV alone. This concentration means inventory and marketing can be highly targeted rather than distributed across the full catalog.

03 RFM Customer Segmentation

SQL TECHNIQUES: `DATE_DIFF` · `NTILE(4)` · Multi-CTE chaining · `CASE WHEN`

Key Finding

A critical retention problem is visible in the data. Over 50% of the customer base is either at risk or already lost, while only 3% qualify as Champions. The Loyal segment (31%) represents the largest opportunity for targeted re-engagement campaigns.

Segment	Customers	Share
Champion	1,948	~3%
Loyal	22,980	~31%
Potential	12,233	~16%
At Risk	24,406	~33%
Lost	12,483	~17%

04 Delivery Performance by State

SQL TECHNIQUES: [DATE_DIFF](#) · [COUNTIF](#) · [Conditional aggregations](#) · [Multi-table JOIN](#)

Key Finding

Delivery performance varies dramatically across Brazil. AL (Alagoas) shows the highest delay rate at 21.4%, compared to just 3.75% in AC (Acre). Northeast states consistently underperform vs. southeast states, pointing to clear logistics infrastructure gaps that directly impact customer satisfaction scores.

05 Delivery Delay vs. Review Score Correlation

SQL TECHNIQUES: [DATE_DIFF](#) · [Bucket classification](#) · [CASE WHEN](#) · [AVG\(\)](#) aggregation

Key Finding

Every additional delay bucket drops the average review score by ~0.7 points. Orders delayed more than 7 days average a score of 1.7 — nearly the worst possible rating. This is not just an operational problem: it is a reputation and retention problem. Improving on-time delivery is the single highest-leverage action to improve customer satisfaction.

Delivery Status	Avg Review Score	Orders
Early	4.30	87,201
On Time	4.10	2,748
1–3 days late	3.29	1,856
4–7 days late	2.10	1,756
+7 days late	1.70	2,798

06 Seller Ranking: Volume vs. Quality

SQL TECHNIQUES: [DENSE_RANK\(\)](#) · [Multi-table JOINs](#) · [AVG\(\)](#) · [COUNT\(\)](#)

Key Finding

6 sellers with 50+ orders have an average review score below 3.0, representing a significant reputational risk. The highest-volume offender ranks #84 overall with 209 orders and \$41K in GMV yet maintains only a 2.71 average review. These sellers generate revenue while systematically degrading the platform's customer experience. A minimum score threshold policy is recommended.